

Same 4 jobs, $N = 2$ nodes, $p(I) = 60\text{s}$. Each block = one 60-second image wait.

Sequential  2.0 job-min (lower bound)

Rolling  3.5 job-min (realistic)

Full burst  4.0 job-min (upper bound)

Sequential: only the first job per node waits. Burst: all jobs see the same cold state. Rolling: middle case.

Half block in *Rolling* = a job that starts mid-pull and waits only the remaining 30 seconds.

Bound: $D_{\text{seq}} = 2.0 \leq D_{\text{roll}} = 3.5 \leq D_{\text{burst}} = 4.0$.